D5.05 — Evolution Learning Tests 46–50: Five Binary Tests for Trust Calibration Currency, Post-Mortem Improvement Quality, DNA Absorption Reasoning Depth, Conflict Carry-Through Completeness, and Vertical Propagation Integrity

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

This note presents five binary operational tests — Tests 46 through 50 — constituting Expansion Category 4: Evolution Learning Tests. Each test addresses a distinct mechanism by which FAI (Full Aspect Integration) events are designed to produce governance learning outputs. The five tests verify: (46) that trust calibration records are current rather than stale; (47) that post-mortem reviews produce specific actionable governance improvement recommendations rather than generic observations; (48) that DNA absorption decisions are documented with substantive governance reasoning; (49) that preserved conflicts carry through as annotations into the home action layers of all participating Selves; and (50) that FAI-origin provenance is maintained as action-layer records propagate upward through the governance hierarchy. Collectively, these tests distinguish governance structures that exist from governance structures that are functioning — the distinction between having an evolution learning process and having one that actually produces learning.

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1. The output-verification framing for evolution learning tests

FAI events are designed to produce learning. Each event generates action-layer records that feed into each participating Self's action-feedback evolution mechanism. Each event may generate DNA-layer content eligible for absorption into home DNA under governance-authorized directed selection. Preserved conflicts annotate governance boundaries that home governance must address. Post-mortem reviews extract improvement commitments. Trust calibration records maintain current assessment of each FAI partner relationship.

These are not administrative processes bolted onto FAI governance — they are the mechanisms by which FAI events produce durable governance improvement. The governance architecture specifies that all five mechanisms exist. The evolution learning tests in this category verify something different and more demanding: that the mechanisms are operating and producing their intended outputs.

The distinction matters because structural compliance and output production can diverge. A governance deployment may maintain a trust calibration process (structurally compliant) while allowing calibration records to become stale (output failure). A post-mortem review process may exist and be completed (structurally compliant) while producing only generic observations rather than actionable recommendations (output failure). Carry-through annotations may be specified in the conflict registry (structurally compliant) while failing to appear in home action layers (output failure). Tests 46–50 are designed to detect exactly this divergence: governance that has the structure but not the function.

2. Test 46 — Trust Calibration Currency

Records to examine: Governance trust calibration records for each active FAI partner; trust calibration review cadence specification.

Binary question: For each active FAI partner relationship, does a governance trust calibration record exist that has been updated within the configured review cadence — that is, the most recent update does not exceed the cadence interval from the current evaluation date?

PASS indicates: A current trust calibration record exists for each active FAI partner, and the most recent update falls within the configured cadence interval. The Self is making FAI governance decisions — including which aspects to contribute, what persistence policies to configure, and how to calibrate sharing scope — on the basis of a current trust assessment of each partner relationship.

FAIL indicates: A calibration record is absent for one or more active FAI partner relationships, or the most recent update for a record exceeds the cadence interval. Trust calibration is stale. The Self is configuring FAI governance using an outdated assessment of a partner relationship. This failure is significant because trust calibration is not merely a documentation exercise — it is the substrate-level record that informs governance configuration for FAI events with that partner. Stale calibration leaves governance misconfigured relative to the current state of the relationship.

The cadence specification is itself substrate content per the configuration-as-substrate-content commitment. The test is therefore binary relative to that content: either the cadence is being met or it is not. The test does not evaluate the depth or quality of calibration reasoning — only whether the cadence commitment is being honored as the precondition for all other trust-sensitive governance decisions.

3. Test 47 — Post-Mortem Governance Improvement Quality

Records to examine: Post-mortem review record for the FAI event under evaluation.

Binary question: Does the post-mortem record contain at least one specific, actionable governance improvement recommendation — identified by governance commitment and proposed action — rather than only general observations?

PASS indicates: The post-mortem contains at least one improvement recommendation that names a specific governance commitment (such as a sharing-scope configuration, a persistence policy, or an orchestration rule) and proposes a concrete action to address it. The post-mortem process is producing its intended learning output: a durable, actionable signal that can feed governance improvement proposals at the appropriate level.

FAIL indicates: The post-mortem record is absent, contains only a generic conclusion ("governance was satisfactory" or equivalent), or contains only descriptive observations without actionable recommendations. The post-mortem process is not producing its intended output. FAI events are occurring and governance decisions are being made, but the learning extraction step is failing. Patterns of governance friction that recur across FAI events will not be detected or addressed through this mechanism. The governance improvement channel that post-mortem review is designed to supply will remain empty regardless of how many FAI events occur.

The test applies a minimal sufficiency threshold — at least one specific actionable recommendation. It does not require that all governance issues from the event be captured, only that the post-mortem process is producing outputs of the right form. A post-mortem that meets this threshold may still be incomplete; a post-mortem that fails this threshold is not functioning as a learning mechanism regardless of its length or formality.

4. Test 48 — DNA Absorption Governance Depth

Records to examine: DNA absorption review records (per the review step) and absorption decision records (per the decision step) for each candidate reviewed within the evaluation period.

Binary question: For each DNA absorption candidate reviewed, does the record document the governance reasoning for the absorption decision — whether absorbed or declined — including what specifically about the candidate informed the decision?

PASS indicates: Each absorption or decline decision is documented with substantive reasoning that references the candidate's content and its fit (or lack of fit) with home governance. The DNA absorption process is operating as a governed selection process, not an administrative check. Home governance can demonstrate why each decision was made, and the absorption record serves as a learning artifact for future absorption decisions — the deployment accumulates a governance-reasoned record of what it considers appropriate content for its DNA layer.

FAIL indicates: Absorption or decline decisions are recorded without documented reasoning. Governance cannot demonstrate the basis for each decision. More critically, the learning value of absorption decisions — the accumulation of a record of what home governance considers a good fit for its DNA layer — is not being captured. DNA evolution from FAI is occurring but is not governed at the reasoning level. The decisions are being made and executed, but they are not building the governance knowledge base that makes future decisions more considered.

The test applies at the individual-decision level. A deployment that documents reasoning for absorbed candidates but not for declined candidates fails this test. Governance reasoning for declines is equally important learning content — it defines the boundary of what home governance considers appropriate DNA content, which is precisely the judgment that governed selection requires.

5. Test 49 — Conflict Carry-Through Annotation Completeness

Records to examine: Conflict registry (PRESERVED entries); participating Selves' home action-layer records following FAI dissolution.

Binary question: For each PRESERVED conflict in the conflict registry, does the carry-through annotation appear in the home action layers of both participating Selves — or all participating Selves for events with more than two participants?

PASS indicates: Carry-through annotations are present in all participating Selves' home action layers for all PRESERVED conflicts. The primary learning mechanism for preserved conflicts is operating: governance boundary signals from FAI are reaching home governance, where they can feed action-feedback evolution and trigger governance improvement proposals at the appropriate level.

FAIL indicates: One or more PRESERVED conflicts do not have corresponding carry-through annotations in all participating Selves' home action layers. This is a high-value failure.

The preserve tier of the three-tier conflict-handling mechanism is designed to surface genuine governance boundary disagreements — cases where automated resolution via orchestration rules was insufficient and escalation to humans across joint authority was not triggered. A preserved conflict is a signal that governance across the participating Selves encountered a genuine boundary condition during the FAI event. That signal is intended to travel back to home governance via the carry-through annotation, where it enters home action-feedback evolution as a record that governance encountered a limit it has not yet resolved.

When carry-through annotations are absent, the most important learning output from competition FAI events and genuine governance conflicts is not reaching the systems that need to act on it. The intelligence about where governance boundaries lie is generated by the FAI event but not transmitted home. The preserved conflict exists in the shared substrate record, but it does not enter the learning machinery of either Self's home governance. FAI events accumulate without building the home governance knowledge about inter-Self boundary conditions that the preserve tier is specifically designed to supply.

The test is bilateral — or n-lateral for N>2 participants. The annotation must appear in ALL participating Selves' home action layers, not only one. A preserved conflict is a cross-perimeter event; its carry-through annotation must cross the same perimeters in the return direction. A partial carry-through — present in one Self's home action layer but absent from another's — fails the test, because the learning mechanism is operating asymmetrically across an event that was symmetric.

6. Test 50 — Vertical Propagation Integrity for FAI-Origin Records

Records to examine: FAI-origin action-layer records in the home substrate; improvement proposals at aspect or Self level that cite those records.

Binary question: For FAI-origin records present in the home action layer, can their FAI-origin provenance be traced through at least one level of vertical propagation — from the level of entry into the home substrate to the next tier in the governance hierarchy?

PASS indicates: FAI-origin provenance is maintained in improvement proposals that propagate upward from FAI-origin action-layer records. Inter-Self learning is traceable through the governance hierarchy. Governance operators reviewing proposals at higher tiers can identify which proposals derive from FAI events, enabling them to assess inter-Self learning patterns, weight proposals accordingly, and maintain accountability for how inter-Self exchange is feeding governance improvement across the deployment.

FAIL indicates: FAI-origin provenance is absent from upward-propagating proposals. This is the failure pattern named AP-15 (Vertical Propagation Break): records that entered the home substrate from FAI events have lost their provenance marker as they move up the governance hierarchy. The practical consequence is that improvement proposals reaching senior governance cannot be identified as FAI-derived, and the governance hierarchy cannot distinguish internally-generated improvement proposals from those that originated through inter-Self exchange. Inter-Self learning continues to flow, but it is not traceable, and governance cannot take its FAI origin into account differentially.

This test is the second stage of a two-stage provenance integrity assessment. The first stage — whether FAI-origin records in the home action layer carry provenance markers at all at the point of entry — is addressed in the AP-15 detection test in D5.03. Test 50 asks whether that provenance survives vertical propagation upward through the governance hierarchy. The two tests compose into a complete lifecycle assessment: provenance acquired at the home action layer (D5.03), and provenance maintained through governance hierarchy propagation (Test 50). A deployment can pass the D5.03 test while failing Test 50: FAI-origin provenance is correctly assigned when records enter the home substrate but is stripped or lost as those records feed upward-propagating improvement proposals. The combination of both tests is required to verify the full provenance lifecycle for FAI-derived governance learning.

7. The five tests as a governance output verification suite

Tests 46–50 share a structural logic: each addresses a mechanism that governance deploys in order to learn from FAI events, and each asks whether that mechanism is producing its intended output rather than merely existing as an architectural specification.

The failure modes they detect are distinct and non-overlapping. Stale trust calibration (Test 46 FAIL) means the Self's governance configurations for FAI are running on outdated partner assessment. Generic post-mortems (Test 47 FAIL) mean FAI events are not generating the improvement recommendations they are designed to generate. Undocumented absorption decisions (Test 48 FAIL) mean DNA evolution from FAI is occurring without a governance reasoning record. Missing carry-through annotations (Test 49 FAIL) mean the most valuable intelligence from preserved conflicts — genuine governance boundary signals that the preserve tier specifically exists to surface — is not reaching home governance. Lost FAI-origin provenance (Test 50 FAIL) means inter-Self learning cannot be traced through the governance hierarchy even when it is present and active.

A governance deployment may pass all anti-pattern detection tests and all composition integrity tests while failing multiple tests in this category. The two categories are not redundant: anti-pattern tests verify structural properties of the governance architecture; evolution learning tests verify that the architecture is functioning and producing the learning outputs it is designed to produce. Both are necessary for a complete operational audit of FAI governance. A deployment that passes structural tests but fails output tests has built the machinery of learning without operating it.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model.* April 2026. ORCID: 0009-0004-8065-3235.

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